

EXECUTIVE REPORT

Supply Chain Performance Report

Concise executive analysis of commercial performance, delivery reliability, profitability, and predictive delay risk across a global sporting goods supply chain.

Source basis: dashboard narrative and EDA summary shared in this conversation | Analysis window: Jan 2015–Jan 2018 | Dataset: 172,765 orders

TOTAL SALES

\$35.21M

Healthy demand across regions and categories

ORDERS

173K

High fulfillment workload

LATE DELIVERY RATE

54.71%

More than half of all orders arrive late

PROFIT AT RISK

\$2.1M

Profit exposed by delayed shipments

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1. Overview Summary

The business is commercially healthy but operationally fragile. It generated **\$35.21M in sales** from roughly **173K orders** and produced **\$3.81M net profit**. However, delivery performance is the dominant weakness: **54.71% of shipments were late**, making delay the default customer experience rather than an exception.

Critical finding: the organization remains profitable today, but logistics underperformance threatens customer trust, premium-shipping credibility, and roughly **\$2.1M of profit** exposed to refunds, compensation, churn, and rising service cost.

- **Demand is strong:** transaction volume is sufficient to support multiple regions and departments.
- **Revenue concentration is high:** Fan Shop contributes almost half of department revenue; Fishing is the strongest sales and profit category.
- **Operational stress is visible:** high-sales markets such as Western Europe also show elevated delivery risk.
- **Customer value is uneven:** Consumer segment profit is strongest; Home Office contributes least.

2. Business Context and Objectives

The company sells sporting goods, outdoor gear, fitness equipment, footwear, and apparel across international markets. The core business problem is a consistent gap between promised and actual delivery performance.

Dataset: 172,765 orders

Period: Jan 2015–Jan 2018

Commercially profitable

Operationally unstable

Objectives

- Measure the severity of late delivery and its impact on service reliability.
- Identify operational bottlenecks causing shipment delays.
- Quantify the financial effect of delay on profit quality and loss-making orders.
- Understand where problems are localized versus company-wide.
- Use machine learning to predict likely delayed shipments before dispatch.

3. KPIs

TOTAL SALES \$35.21M Strong revenue base across the global network	TOTAL ORDERS 172,765 Large sample and high operational throughput	NET PROFIT \$3.81M Positive margin despite logistics issues
AVERAGE ORDER VALUE \$203.83 Mid-sized basket; cross-sell opportunity remains	AVERAGE MARGIN 12% Reasonable, but not resilient to avoidable friction	ACTUAL VS SCHEDULED 3.50 vs 2.93 days Average delay is small, but spread across most orders

KPI	Finding	Executive reading
Late orders	94,523–95K	Scale of delay is large enough to affect brand trust and cost structure.
On-time orders	~78K	On-time performance is materially below mature supply chain expectations.
Shipping mode mix	Standard ~60%, Second ~20%, First ~15%	Standard Class dominates volume, but premium modes are the bigger reliability failure.
Commercial concentration	Fan Shop and Fishing lead	Top-line performance depends on a narrow set of winners, increasing concentration risk.

4. Profitability

Most orders are profitable, but profit quality is weakened by delivery issues and concentration in a small number of categories and departments.

80.7% profitable

18.7% loss-making

0.6% neutral

- **Profit at risk:** delayed shipments account for approximately **\$2.1M** of profit exposed to service failure costs.
- **Delay-cost profile:** short delays are frequent; longer delays become progressively more expensive on an average-profit basis.
- **Category concentration:** Fishing is both the highest-selling and highest-profit category; Cleats are second, while several categories contribute far less.
- **Department concentration:** Fan Shop is the largest profit and revenue contributor; Apparel follows; Technology and Outdoors remain weak.
- **Regional value:** Western Europe is the strongest revenue/profit region, with Central America also important.

Implication: the business is not only exposed to delay risk, but also to concentration risk. Protecting Fishing supply continuity and diversifying beyond Fan Shop should be treated as a margin-protection initiative, not only a growth initiative.

5. Bottleneck Distribution

Shipping mode is the clearest operational bottleneck. Regional hotspots exist, but the pattern of delay is broad enough to indicate a company-wide fulfillment problem rather than a single bad market.

Area	Observed pattern	Interpretation
First Class	100% delayed	Premium promise is not being met at all; urgent management priority.
Second Class	79.8% delayed	Expedited shipping is materially underperforming and damaging service credibility.
Standard Class	39.8% delayed	Still too high, but materially better than premium modes.
Same Day	0% delayed	Suggests promise-setting and process control matter more than speed alone.
Regions	Western Europe and Southeast Asia appear as hotspots in dashboard views	Hotspots exist, but EDA indicates delay rates are broadly similar across regions; the issue remains company-wide.
Departments / categories	Health & Beauty, Pet Shop, Golf Bags, TV, Soccer, Racquet, Pet Supplies show added friction	Handling complexity and inventory imbalance appear to be secondary contributors.

6. Root Cause Analysis

1. **Premium shipping overpromise:** First and Second Class perform worse than Standard, indicating unrealistic SLAs, carrier mismatch, or poor capacity allocation.
2. **Systemic process inefficiency:** most delays cluster in the **0–2 day range**; this points to frequent small execution misses rather than rare catastrophic failures.

3. **Fulfillment planning weakness:** similar delay rates across customer segments and many regions suggest the core process is broadly underperforming.
4. **Payment and release friction:** PAYMENT_REVIEW and PENDING_PAYMENT orders show elevated delay, implying upstream order-release bottlenecks.
5. **Capacity strain in high-demand periods and products:** peak months and specialty items show added delay risk, consistent with warehouse congestion and handling complexity.

Root-cause conclusion: the late-delivery issue is primarily a planning and execution problem — especially in premium shipping design — not a one-region infrastructure collapse.

7. Time-Based Delay Pattern

- **Peak delay months:** **August**, **September**, and **December** show the highest late-delivery rates, consistent with predictable seasonal surges.
- **Weekday effect:** Monday is only slightly worse than Sunday; the difference across the week is under 1%, so the problem is not tied to one day.
- **High-risk hours:** **9 PM**, **11 AM**, and **12 PM** show the highest delay risk, pointing to cut-off-time misses and midday warehouse congestion.

Executive reading: the company is underprepared for predictable demand spikes and specific processing windows. Better seasonal staffing, workload balancing, and order cut-off rules could remove a large share of delay without major network redesign.

8. ML Model Results

The updated Random Forest model is operationally useful for pre-shipping risk detection and should replace the outdated 74% summary in prior draft text.

Accuracy: 0.75

Precision: 0.84

Recall: 0.70

Test support: 34,553

Class	Precision	Recall	F1-score	Support
0 = non-delayed	0.67	0.82	0.74	14,758
1 = delayed	0.84	0.70	0.76	19,795

- **Precision 0.84:** when the model flags an order as likely late, it is correct **84% of the time**. This is strong for prioritizing intervention.
- **Recall 0.70:** the model identifies **70% of all actual late shipments**, but about 30% still go unflagged.
- **Business use:** apply the model immediately after order confirmation to trigger expedited handling, proactive customer messaging, or alternative carrier selection.

9. Recommendations

Priority 1 — Repair premium shipping reliability

Audit First Class and Second Class end-to-end, including carrier performance, promised transit times, capacity allocation, and exception handling. If SLAs cannot be met consistently, reprice or temporarily limit premium options.

Priority 2 — Deploy predictive delay alerts

Integrate the Random Forest model into the order workflow. Flag high-risk orders before dispatch, route them for faster handling, and notify customers early when delay risk is elevated.

Priority 3 — Fix upstream fulfillment release bottlenecks

Automate payment validation where possible, escalate payment reviews faster, and shorten the time between order approval and warehouse release.

Priority 4 — Prepare for predictable demand spikes

Build capacity plans for August, September, and December; review late-evening order cut-off rules; add labor or carrier capacity around midday processing congestion.

Priority 5 — Protect core profit drivers while diversifying

Safeguard Fishing inventory and supplier continuity, continue investing in Western Europe and other strong markets, and grow lower-performing departments to reduce dependence on Fan Shop and a narrow product set.

10. Conclusion

The business has strong commercial momentum, but its logistics model is not keeping pace. Revenue, demand, and net profit remain healthy, yet **54.71% late delivery** means customer experience is being systematically degraded. The main problem is not demand weakness — it is premium shipping failure, small but frequent fulfillment delays, and insufficient preparation for predictable operational peaks.

Near-term improvement should focus on shipping-mode redesign, operational release speed, seasonal capacity planning, and production use of the updated predictive model. If leadership acts on these areas, the company can improve on-time performance, protect the **\$2.1M profit at risk**, and reduce dependence on reactive service recovery.

Target KPI	Current	Target	Business impact
Late delivery rate	54.71%	<30% in 12 months	Stronger customer satisfaction and lower service cost
Premium shipping reliability	First: 100% delayed Second: 79.8% delayed	Restore premium on-time credibility	Protect brand trust and justify premium pricing
Profit at risk	\$2.1M	Reduce by 40%	Improve margin quality and reduce compensation leakage
Predictive model effectiveness	Accuracy 0.75 Recall 0.70	Higher recall with stable precision	Catch more risky orders before shipment

Note: This report intentionally uses the updated Random Forest metrics supplied by the user (Accuracy 0.75, Precision 0.84, Recall 0.70) in place of older draft figures.